SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –SIH26003**
- **Problem Statement Title- AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Software**
- **Team ID-1**
- **Team Name (Registered on portal)- Sidequest**

SideQuest

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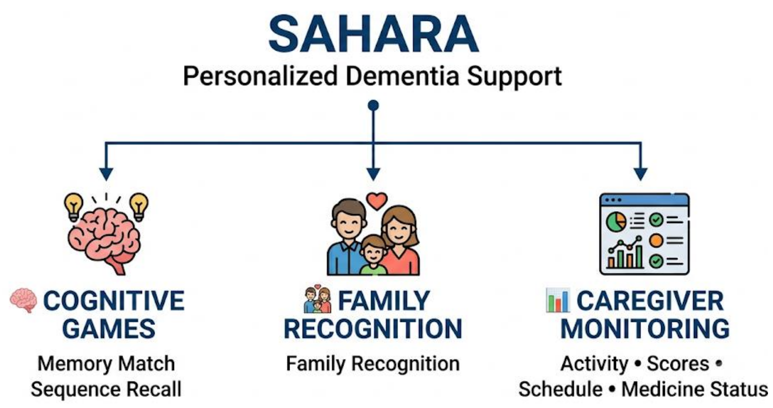
SAHARA – Smart Dementia Support & Cognitive Gaming Platform

Proposed Solution
SAHARA is a dementia-support application with separate **Patient** and **Caretaker** interfaces.
Patient: 3 cognitive games — **Memory Match, Sequence Recall & Family Recognition**.
Caretaker: Monitors **activity, scores, schedules, medicine status & family photos**.
Accessibility: Voice-assisted instructions and language support.



How it addresses the problem
Provides **simple, engaging and personalized cognitive activities** while helping caregivers monitor patient activity and performance.

Innovation
Combines **cognitive gaming + family recognition + caregiver monitoring**, with **randomized content and accessibility features** in one platform.

TECHNICAL APPROACH

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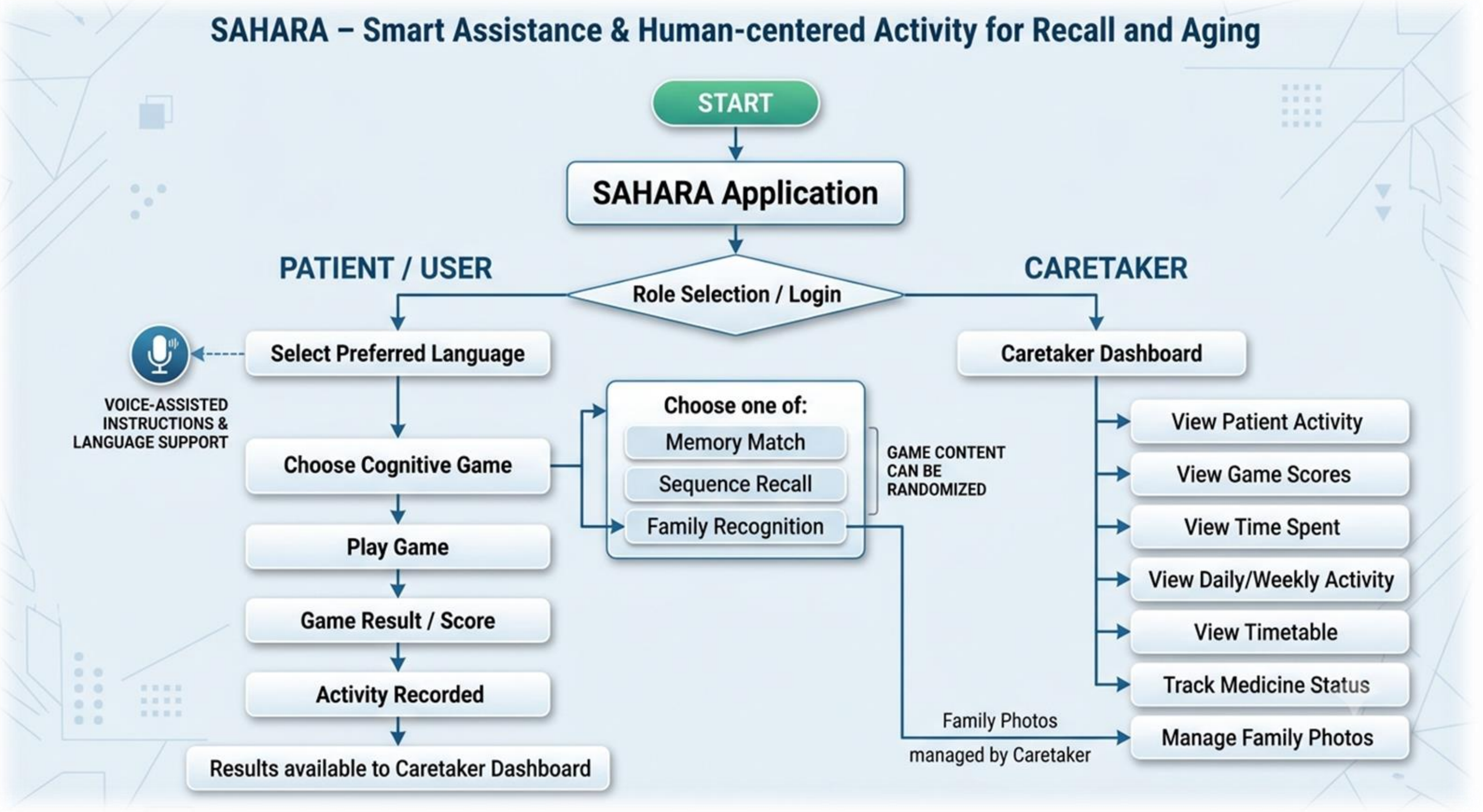
<u>Technologies used</u>

HTML + CSS + TypeScript→ Frontend
Local Storage / Database → Data
Speech Technology → Voice
Role-Based Authentication → Security

<u>Process Flowchart</u>



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FEASIBILITY AND VIABILITY

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IMPACT AND BENEFITS

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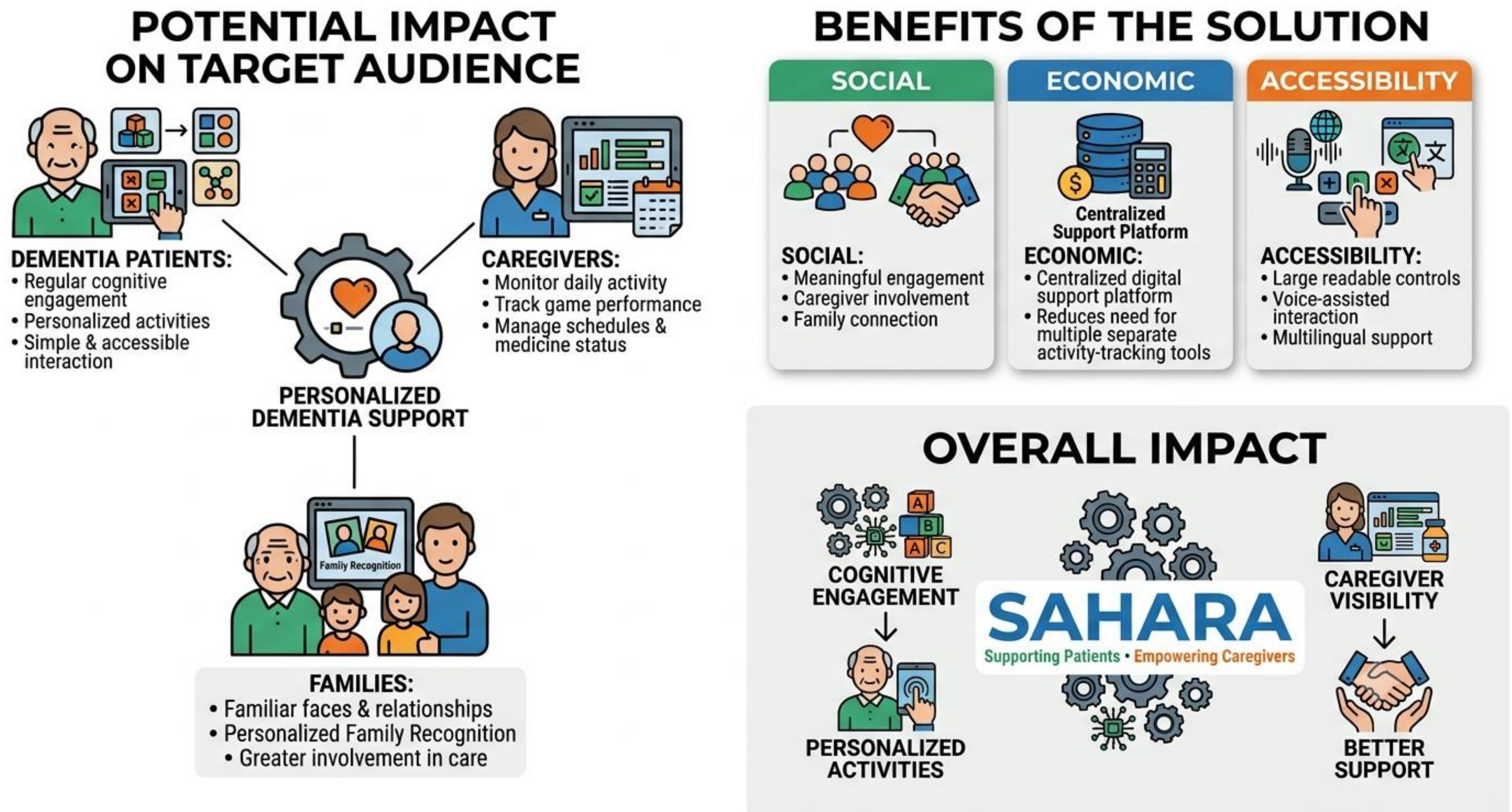

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RESEARCH AND REFERENCES

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- **Details / Links of the reference and research work**
- Effectiveness of Internet-Based or Mobile App Interventions for Family Caregivers of Older Adults with Dementia: A Systematic Review – PubMed : Internet and mobile-based interventions can improve the well-being and support needs of dementia caregivers.
- Home-Based Digital Healthcare Interventions for Dementia: A Systematic Review of Patient and Family Caregiver Outcomes - PubMed :Digital, personalized, home-based interventions can support cognitive health in dementia while reducing caregiver burden and stress.